



SWMS08

Safe Work Method Statement

Work in or near a tunnel (ecological survey activities)

Document number	EC-SWMS08	Revision	1.0
Prepared by	Ecology Consulting — WHS / Field team	Document release date	
Approved by	Tony Webster — Managing Director / WHS Officer	Next review	

Note: Work must be performed in accordance with this Safe Work Method Statement (SWMS). This SWMS must be kept and available for inspection until the work is completed. If revised, every version must be kept. If a notifiable incident occurs in relation to the work in this SWMS, the SWMS must be kept for at least two years from the date of the notifiable incident. This SWMS is accessed and acknowledged through the Ecology Consulting digital WHS platform; field forms are submitted from site and reviewed centrally by the administration team.

Scope & limitations: Ecology Consulting personnel will not enter confined spaces as defined under WHS Regulations unless a confined space entry permit system is implemented and authorised by the Principal Contractor. Works are limited to ecological observation and survey only. Personnel will not undertake construction activities, confined space entry (unless authorised), or interfere with plant, services or structural elements. Entry proceeds only where the tunnel is assessed as *not* a confined space, or where entry is controlled under a confined space permit system — see SWMS06.

Project & PCBU details

Person conducting a business or undertaking (PCBU)	Ecology Consulting Pty Ltd · ABN 68 647 633 163	Contact number	0472 569 988 (office / Check-in)
EC offices	34 Atchison St, Wollongong NSW 2500 · 42 Goulburn St, Crookwell NSW 2583 · 473 Townsend St, Albury NSW 2640		
Client / Principal Contractor			
Work activity	Ecological observation and survey in or near tunnels, culverts and underpasses, including fauna assessment, habitat inspection and device deployment and retrieval.		
Workplace location			

Field work duration	
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Legislation, codes of practice & standards

Acts & Regulations	Work Health and Safety Act 2011 (NSW) and the Work Health and Safety Regulation (NSW); Work Health and Safety Act 2011 (Cth) and WHS Regulations 2011 (Cth) where work is on Commonwealth land or for a Commonwealth entity (Comcare jurisdiction); Work Health and Safety Act 2011 (ACT) and WHS Regulation 2011 (ACT) for works in the ACT.
Codes of Practice	Safe Work Australia model Codes as adopted in NSW and the ACT: <i>How to manage work health and safety risks; Work health and safety consultation, cooperation and coordination; Construction work; First aid in the workplace; Managing the work environment and facilities; Preparation of safe work method statements for construction work. Confined spaces; Managing the risk of falls at workplaces; Managing risks of plant in the workplace.</i>
Standards & guidance	AS 2865 Confined spaces; AS/NZS 1715 and AS/NZS 1716 Respiratory protective equipment; AS/NZS 1801 Occupational protective helmets; AS 1319 Safety signs.
EC WHS documentation	EC Work Health and Safety Management System and WHS Handbook; EC Risk Management Processes (EC-RMP-FRM-001); EC Hazard Register and Trigger Action Response List (TARL); Pre-Mobilisation Checklist; Pre-Travel Vehicle Safety Checklist; Journey Management Plan; Daily Toolbox Talk; site-specific Emergency Response Plan (ERP); JSA — General Field Surveys; JSA — Remote and Isolated Work; SOP — Journey Management; SOP — Terrestrial Fauna; SOP — Fieldwork Biosecurity.

Mandatory personal protective equipment & field equipment

Confirmed present and serviceable on the Pre-Mobilisation Checklist and verified by the Field Lead at the Daily Toolbox Talk. Faulty items are tagged out and removed from service.

Hard Hat	P2 Respirator	Safety Glasses	Gloves	Hearing Protection	Hi-Vis Clothing	Disposable Coveralls	Long Sleeves	Capped Boots
☐	☐	☐	☐	☐	☐	☐	☐	☐

Fire Extinguisher	Remote First Aid Kit	Snake Bite Kit	In Reach / Sat Phone	Two-Way Radio	Torch / Headlamp	GPS	Drinking Water
☐	☐	☐	☐	☐	☐	☐	☐

Gas Detector	Torch (two per person)	Spare Batteries	Backup Comms Device	Tunnel Safety Plan (PC)	Entry Authorisation	Exclusion Zone Signage	Two-Person Team
☐	☐	☐	☐	☐	☐	☐	☐

Hierarchy of controls

Eliminate	Avoid entering tunnels or culverts where practicable — conduct inspections from the entrance, and use cameras, torches or remote survey methods.
Substitute	Use safer survey approaches — visual assessment from outside, non-entry fauna detection methods, and remote lighting and monitoring equipment.

Isolate	Restrict access to unsafe tunnels or sections; establish exclusion zones where structural, air quality or flooding risks are present; maintain safe distances from hazards.
Engineer	Use available controls such as fixed access points, existing walkways, lighting systems and ventilation provided by the Principal Contractor.
Administrative	SWMS, minimum team requirements, communication protocols, emergency plans, air quality verification, entry permissions and a documented go / no-go assessment; enforce stop-work triggers.
PPE	Last line of defence — helmets, headlamps, gloves, appropriate footwear and respiratory protection where required.

Hazards & controls

A documented go / no-go assessment must be completed and all conditions verified safe before any entry.

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Planning, preparation and travelling to site		
Inadequate planning leading to uncontrolled risk; unfit or untrained personnel	Attend the pre-field meeting with the Project Lead and Project Manager to review scope, methodology and expected hazards; confirm site access, keys, codes, inductions and property constraints; and confirm all staff are trained, competent and fit for work. Medical concerns are disclosed privately to the Field Lead. Review this SWMS, the relevant JSA and SOP, the Field Plan and the site-specific ERP. Verify training currency — workers are not mobilised with expired mandatory training.	☐ Pre-field meeting ☐ SWMS / JSA / SOP reviewed ☐ ERP reviewed ☐ Training verified ☐ Medical disclosures
Vehicle failure, missing safety equipment or fatigue during travel	Complete the Pre-Travel Vehicle Safety Checklist — first aid kit, extinguisher, spare tyre, torch, water and PPE present; vehicle free from leaks, faults or damage; electrical issues visually inspected only and tagged out if unsafe. Complete the Journey Management Plan. Minimum 10-hour rest between shifts before driving; personal fatigue self-assessment before departure; rest breaks every two hours; carry water, snacks and electrolytes; second driver for long or remote trips where practicable. Zero blood alcohol concentration applies to all work driving.	☐ Vehicle checklist done ☐ JMP completed ☐ 10-hr rest confirmed ☐ Fatigue self-assessment
Travelling into fire, flood, storm, smoke or hazardous road conditions	Complete the Pre-Travel Environmental and Hazard Check using BOM weather warnings, Hazards Near Me, RFS fire danger ratings, Air Quality Index, lightning trackers, Live Traffic NSW and telecommunications coverage maps. Do not travel where there are emergency warning fire events, hazardous air quality, active storm warnings or flooded roads. Never drive through floodwater under any circumstance. If conditions change, stop, reassess and notify the Communication Officer.	☐ BOM / Hazards Near Me ☐ RFS rating checked ☐ AQI checked ☐ Live Traffic NSW ☐ Coverage checked
Hazards not identified, assessed or communicated before work commences	All team members participate in a pre-commencement Daily Toolbox Talk led on site by the Field Lead and documented on the EC WHS form — site-specific hazards identified; collaborative risk assessment using the EC risk matrix (rating one to six) with agreed controls and a revised residual rating; PPE and equipment readiness confirmed; ERP reviewed and Emergency Meeting Point identified; communication and check-in requirements confirmed; health or access needs disclosed privately; and worker sign-on confirming understanding and fitness for work. Hazards already in the EC Hazard Register are managed under the TARL. No worker commences without sign-on.	☐ Toolbox talk held ☐ Risk assessment rated ☐ Emergency Meeting Point ☐ All workers signed on
Loss of contact with the field team; delayed emergency response	Mandatory check-ins with the Communication Officer (0472 569 988) before departing base, on arrival at site or accommodation, before leaving site, and on return to base. Every check-in must be acknowledged. A missed or unacknowledged check-in triggers escalation through the Communication Officer, Project Lead or Project Manager, and the WHS Officer, and is grounds for activating the ERP. Carry redundant communications — mobile phone, UHF radio and In Reach or satellite communicator, with a PLB where remoteness warrants. Roles are assigned at the toolbox talk: Field Lead, Communication Officer and workers.	☐ Check-in schedule agreed ☐ Comms tested ☐ Roles assigned ☐ Escalation briefed
Planning, consultation and pre-start		

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Unknown hazards within the tunnel or culvert — structural instability, contaminants, confined space conditions; incomplete or inaccurate information from the Principal Contractor	Obtain and review all available site information prior to entry. Confirm the tunnel classification — confined space or non-confined space — in writing. Confirm Principal Contractor controls are implemented and effective. Treat the tunnel as hazardous if information is incomplete and do not proceed without confirmation of safe conditions.	☐ Site information reviewed ☐ Classification confirmed ☐ PC controls verified ☐ Confirmation received
Lack of emergency plan; communication failure	Confirm the Principal Contractor has a tunnel safety plan. Ensure communication systems are in place and tested, including two-way radios and confirmation of mobile coverage. Identify emergency exits and muster points before entry and brief them to the whole team.	☐ Tunnel safety plan confirmed ☐ Comms tested ☐ Exits identified ☐ Muster point briefed
Flash flooding or sudden water flow; entrapment due to upstream rainfall	Review weather forecasts and upstream catchment conditions before entry. Do not enter during or following rainfall events where flow risk exists. Establish a trigger for immediate evacuation if water levels rise, and monitor levels continuously during the work.	☐ Weather checked ☐ Flood risk assessed ☐ Evacuation trigger set
Entry decision, team and isolation		
Entry into an unsafe or uncontrolled environment	Complete a go / no-go assessment prior to entry considering air quality, structural integrity, water risk, lighting and communication. Do not enter unless all conditions are verified safe. Record the outcome and the person authorising entry.	☐ Go / no-go confirmed ☐ All conditions verified ☐ Authorisation recorded
Lone worker risk; delayed emergency response	Minimum two-person team at all times. Assign an On-Ground Communications Officer who remains outside and maintains continuous communication with personnel inside the tunnel. Roles are assigned and acknowledged at the toolbox talk.	☐ Two-person team ☐ Comms Officer assigned ☐ Roles acknowledged
Unauthorised activation of plant; exposure to operating machinery	Confirm lock-out and tag-out systems are in place where required. Do not interfere with plant or isolation systems. Maintain exclusion distance from machinery and confirm with the Principal Contractor that plant will not be operated during the survey.	☐ LOTO verified ☐ No interference ☐ Exclusion distance kept
Working within the tunnel		
Low oxygen, toxic gases, dust and airborne contaminants	The atmosphere must be tested and confirmed safe prior to entry. Use gas monitoring where required and maintain it during the work. Do not rely on assumptions of safe air quality. Exit immediately if an alarm triggers or if any person experiences symptoms such as headache, dizziness or breathlessness.	☐ Air tested ☐ Monitoring active ☐ Exit-on-alarm briefed
Collapse or falling debris; contact with unstable surfaces	Visually inspect for cracks, deformation, spalling or instability before and during entry. Do not enter if structural concerns are identified. Maintain distance from unstable areas and do not touch or load unstable surfaces. Helmets worn at all times.	☐ Structure assessed ☐ No entry if unstable ☐ Distance maintained
Slips, trips and falls on uneven, wet or unstable surfaces; falling from edges or into channels; disorientation in low light	Only enter if authorised and safe to do so. Walk slowly, maintain stable footing and avoid sudden movements. Maintain safe distance from edges and drops. Do not proceed beyond line of sight unless authorised. Maintain constant communication and exit immediately if conditions change.	☐ Entry conditions safe ☐ Footing stable ☐ Line of sight kept ☐ Comms maintained

What are the hazards & risks	What are the control measures	Select relevant controls (completed onsite by Field Lead)
Poor visibility leading to trips or collision; loss of orientation	Carry a minimum of two independent light sources per person with spare batteries. Ensure adequate illumination at all times. Do not proceed if lighting is insufficient.	☐ Two light sources each ☐ Spares carried ☐ Illumination adequate
Loss of contact with personnel; delayed emergency response	Maintain continuous communication with the team and the On-Ground Communications Officer. Use a backup communication system. Exit immediately if communication is lost.	☐ Comms maintained ☐ Backup available ☐ Exit-on-loss briefed
Biological, fauna and hygiene controls		
Guano exposure and associated fungal disease risk; contaminated water or surfaces	Avoid contact with contaminated materials. Where guano or nesting material is present, do not dry-sweep or disturb it; wear a P2 respirator and disposable coveralls. Use gloves and hygiene practices, wash hands before eating or drinking, and decontaminate equipment as required. Report any post-work flu-like symptoms to a medical practitioner and disclose guano exposure.	☐ Contact avoided ☐ RPE worn ☐ Hygiene controls applied ☐ Equipment decontaminated
Bites and scratches during fauna encounter; zoonotic disease exposure	Avoid handling fauna unless authorised and trained. Use appropriate PPE and handling equipment. Follow approved fauna handling procedures in SOP — Terrestrial Fauna. Do not handle bats unless vaccinated and authorised. Withdraw if an animal blocks egress or behaves defensively.	☐ Handling controls applied ☐ Trained handler only ☐ No unvaccinated bat contact
Exposure to contaminants and secondary contamination	Minimum PPE is helmet, boots, gloves and long sleeves, with additional PPE as required including a P2 respirator and coveralls. No eating or drinking in contaminated areas. Remove PPE safely and dispose of it appropriately; do not shake or blow down clothing.	☐ PPE compliant ☐ No eat / drink ☐ PPE removed safely
Working near the tunnel entrance		
Vehicle or plant movement; falling objects or unstable ground at the portal	Maintain awareness of plant and vehicle movement. Stay clear of active work areas. Do not stand under unsupported structures. Use designated safe access paths and maintain the exclusion zone.	☐ Exclusion upheld ☐ Clear of active areas ☐ Designated paths used

Stop-work triggers — stop work immediately, move to a safe location, notify the Field Lead and Principal Contractor, and do not recommence until the risk is controlled and communicated			
Air quality unknown or alarm triggered	Rising water levels or flooding risk	Structural instability observed	Loss of communication
Inadequate lighting	Unidentified hazards present	Plant operating without notification	Any person reports symptoms

Emergency response

Read with the site-specific Emergency Response Plan. The Field Lead is the Chief Warden. Any worker may activate the ERP.

1 · Activate & make safe	Stop work and ensure the area is safe for responders. Do not move an injured person unless there is immediate danger. Rescue is to be conducted by the Principal Contractor or emergency services only — Ecology Consulting staff must not attempt confined space or tunnel rescue . The On-Ground Communications Officer initiates the emergency response, evacuates remaining personnel to the muster point and notifies the Principal Contractor. The Field Lead assesses the situation, initiates the response, directs evacuation to the Emergency Meeting Point, accounts for all personnel using the Daily Toolbox Talk sign-on sheet, and coordinates communication with the office.
2 · Call for help	Call 000 or 112, or activate the In Reach or PLB where there is no coverage. Provide GPS coordinates and the nearest cross roads, the nature of the emergency and safe access routes. Assign one person to guide responders in.
3 · First aid	A trained first aider assesses the casualty, but only where it is safe to do so. Control bleeding, immobilise suspected fractures and monitor breathing. If unconscious but breathing, place in the recovery position; if not breathing, commence CPR. Record all treatment on the First Aid Treatment Log.
4 · Post-incident	Report to the WHS Officer and complete an Incident Report Form within the required timeframe, submitted through the digital WHS platform. The WHS Officer updates the Incident Register and notifies the regulator of any notifiable incident. Attend debrief; review and update this SWMS and the ERP; restock emergency equipment used. Psychosocial support is provided where workers were exposed to distressing events.

Emergency services	000 or 112	State Emergency Service (SES)	13 25 00
EC office / Check-in Officer	0472 569 988	Managing Director / WHS Officer	Tony Webster · 0448 439 334
Poisons Information Hotline	13 11 26	Wildlife rescue	1300 094 737
Employee Assistance Program	1300 172 327 — free and confidential	Nearest hospital & muster point	Identified per site in the ERP; confirmed at the toolbox talk

Training & competency requirements

Required for this activity	Confined space entry training (RIIWHWS202E or equivalent) is required where the tunnel is classified as a confined space or classification is uncertain. All entrants must be briefed on the tunnel safety plan, gas monitoring and emergency procedures prior to entry.
Baseline — all field personnel	White Card (General Construction Induction); current driver's licence of appropriate class; CPR certification (annual renewal); Provide First Aid (HLTAID011), and Remote First Aid (HLTAID013) for remote or isolated work; General Fauna Handling Training.
Verification	All training must be current, recorded and verifiable prior to mobilisation. Records are maintained by the Administration Officer and retained for a minimum of five years, with expiry notifications issued in advance; workers are not mobilised with expired mandatory training. Competency is verified annually and observed on site by Principal or Senior Ecologists, Project Managers and the WHS Officer. Subcontractor training evidence is collected and retained for audit.

Daily SWMS acknowledgement & sign-on register

All personnel must read, understand and sign this SWMS before commencing work, confirming they are fit for duty, competent and able to perform assigned tasks safely. Completed with the Daily Toolbox Talk and submitted through the digital WHS platform.

#	Name	Role	Signature	Date
1				
2				
3				
4				
5				
6				

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